STUDENT DATABASE MANAGEMENT SYSTEM

21CSC205P Database Management Systems
A MINI PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that Project report titled "STUDENT DATABASE MANAGEMENT SYSTEM" is the bonafide work of "SHARVESWAR.M - RA2411003011058, MAGI SHARMA J – RA2411003011045" who carried out the 21CSC205P Database Management Systems mini project work under my supervision.

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ABSTRACT

The Student Database Management System is a web-based application developed to streamline and automate the management of student records in educational institutions. The system uses a **normalized 6-table relational database** to store student data across separate entities: **Students, Guardians, Academic Records, Attendance, Assessments, and Fee Records**. Built using Next.js for the frontend and PostgreSQL for the backend, the application ensures data integrity through **foreign key constraints with cascading deletes** and **transactional writes** across all tables. Key features include a comprehensive student dashboard, multi-table data entry, and a searchable view with expandable student details. The implementation of normalization techniques (up to 3NF) ensures an optimized database design, while transaction management guarantees data consistency during concurrent operations.

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CHAPTER 1

PROBLEM UNDERSTANDING, IDENTIFICATION OF ENTITY AND RELATIONSHIPS, CONSTRUCTION OF DB USING ER MODEL FOR THE STUDENT DATABASE MANAGEMENT SYSTEM

1.1 Introduction

The **Student Database Management System (SDMS)** is a comprehensive software solution designed to streamline the management of student information within an educational institution. In the traditional manual system, maintaining student records, including personal details, academic performance, attendance, and fee payments, is a tedious and error-prone process. This project aims to digitize these operations, providing a centralized platform for administrators, faculty, and students to access and manage data efficiently.

The system is built using modern web technologies, ensuring accessibility, scalability, and security. It facilitates real-time data updates, generates insightful reports, and simplifies administrative tasks such as admission processing and result analysis. By automating routine operations, the SDMS enhances the overall productivity of the institution and ensures data integrity.

1.2 Motivation

The primary motivation behind developing this system is to address the inefficiencies of paper-based record-keeping. Educational institutions handle vast amounts of data, and manual processing often leads to:

- **Data Redundancy:** Multiple copies of the same data across different departments.
- **Inconsistency:** Discrepancies in student records due to manual entry errors.
- **Time Consumption:** Retrieving specific student information takes significant time.
- **Security Risks:** Physical records are susceptible to damage or unauthorized access.

By transitioning to a digital database management system, we aim to eliminate these challenges, ensuring that student data is accurate, secure, and easily retrievable. The motivation is to create a robust system that supports the academic and administrative needs of the college.

1.3 Scope

The scope of the Student Database Management System includes:

- **Student Information Management:** Storing personal, academic, and contact details.
- **Academic Records:** Managing semester-wise marks, GPA, CGPA, and credit hours.
- **Attendance Tracking:** Recording daily attendance and calculating percentages.
- **Fee Management:** Tracking fee payments, dues, and scholarship details.
- **Search and Filter:** enabling quick search of students by ID, name, or department.

- **Data Export:** Facilitating the export of student data for reporting purposes.

The system is designed for use by the college administration and faculty, with potential for future expansion to include a student portal.

1.4 Problem Statement

The current manual system for managing student data is inefficient and prone to errors. Administrators struggle to maintain up-to-date records of hundreds or thousands of students. Tracking fee payments, calculating attendance percentages, and generating academic reports manually is labour-intensive. There is a need for an automated, centralized system that can handle student data securely and provide quick access to information for decision-making.

1.5 Project Requirements

Functional Requirements:

- The system must allow the addition, update, deletion, and viewing of student records.
- It should calculate GPA and CGPA automatically based on marks.
- It must track fee status and highlight students with pending dues.
- It should support data validation to prevent invalid entries (e.g., negative marks).

Non-Functional Requirements:

- **Performance:** The system should respond to queries within 2 seconds.
- **Scalability:** It must handle a growing number of student records without performance degradation.
- **Security:** Access to the database should be restricted to authorized personnel.
- **Usability:** The user interface should be intuitive and easy to navigate.

1.6 Identification of Entity and Relationships

Based on the problem statement, the key entity identified for this phase is the **student**. In a more complex scenario, this would be broken down into multiple entities, but for the initial scope, we are utilizing a comprehensive schema.

Entities:

1. **Student — Core identity and contact information.**
    - *Attributes:* **Student ID (PK), Full Name, DOB, Gender, Address, Phone, Email, Admission Date.**
2. **Guardian — Parent/guardian details.**
    - *Attributes:* **Guardian ID (PK), Student ID (FK), Name, Phone, Email, Relationship.**

3. **Academic Record — Enrollment and academic performance.**
    - ***Attributes:*** **Record ID (PK), Student ID (FK), Enrollment Status, Program, Department, Semester, Credit Hours, GPA, CGPA.**
4. **Attendance — Class attendance tracking.**
    - ***Attributes:*** **Attendance ID (PK), Student ID (FK), Semester, Classes Attended, Total Classes, Percentage.**
5. **Assessment — Examination marks and grades.**
    - ***Attributes:*** **Assessment ID (PK), Student ID (FK), Semester, Internal Marks, Quiz Marks, Semester Marks, Total Marks, Grade.**
6. **Fee Record — Financial tracking.**
    - ***Attributes:*** **Fee ID (PK), Student ID (FK), Total Fees, Fees Paid, Pending Dues, Payment Status, Scholarship Amount, Scholarship Type.**

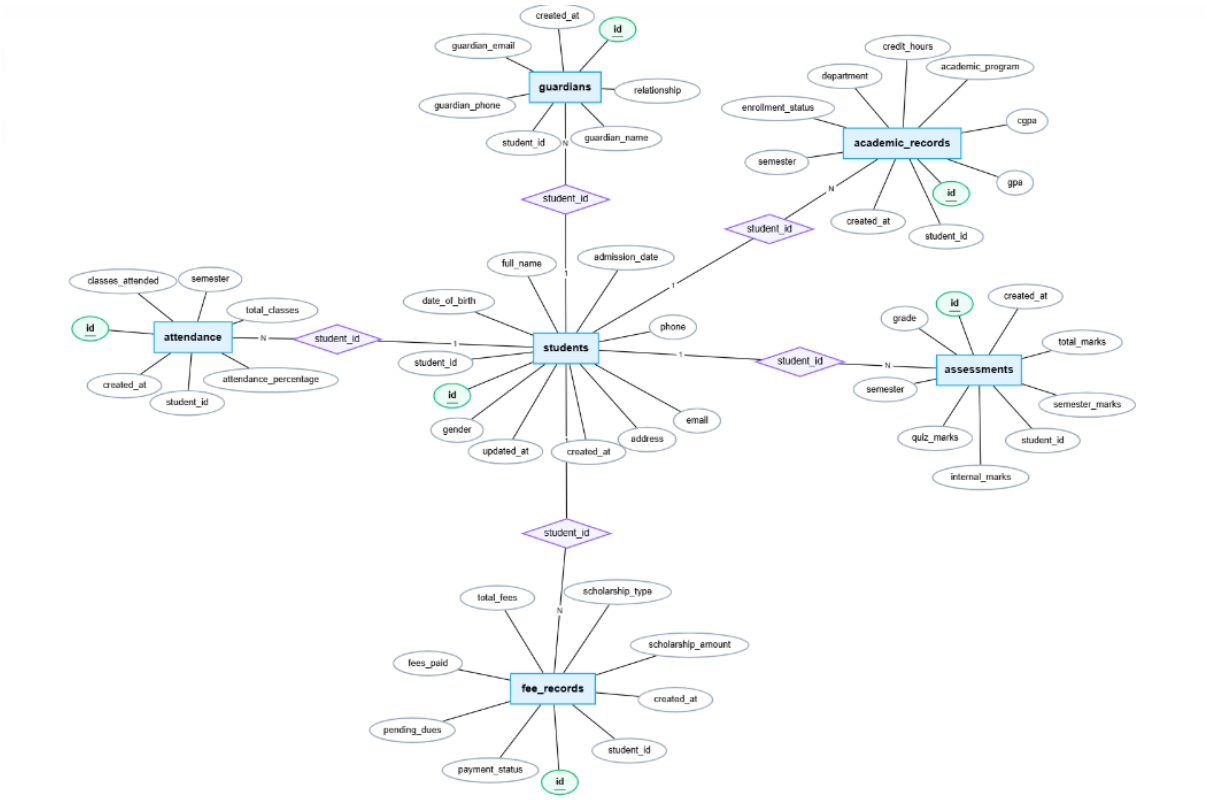
Relationships:

- **A Student has one Guardian (1:1 or 1:N).**
- **A Student has one Academic Record per semester (1:N).**
- **A Student has one Attendance record per semester (1:N).**
- **A Student has one Assessment per semester (1:N).**
- **A Student has one Fee Record (1:1 or 1:N).**
- **All child tables reference students(id) with ON DELETE CASCADE.**

1.7 Construction of DB Using ER Model for the Student Database Management System

The Entity-Relationship (ER) Diagram represents the logical structure of the database.

Figure 1.1: ER Diagram of the Student Database Management System



CHAPTER 2

DESIGN OF RELATIONAL SCHEMAS, CREATION OF DATABASE AND TABLES FOR STUDENT DATABASE MANAGEMENT SYSTEM

2.1 Schema for Student Database Management System

The relational schema for the Student Database Management System is derived from the ER model. Since we are using a comprehensive design, the schema consists of a primary students table that encapsulates all student-related data.

Schema Diagram:

Figure 2.1: Schema for Student Database Management System

dbms > lib > db > schema.sql

```sql
-- 1. Core Student Identity & Contact Information
CREATE TABLE IF NOT EXISTS students (
  id SERIAL PRIMARY KEY,
  student_id VARCHAR(50) UNIQUE NOT NULL,
  full_name VARCHAR(255) NOT NULL,
  date_of_birth DATE,
  gender VARCHAR(20),
  address TEXT,
  phone VARCHAR(20),
  email VARCHAR(255),
  admission_date DATE DEFAULT CURRENT_DATE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Guardian / Parent Information
CREATE TABLE IF NOT EXISTS guardians (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  guardian_name VARCHAR(255) NOT NULL,
  guardian_phone VARCHAR(20),
  guardian_email VARCHAR(255),
  relationship VARCHAR(50) DEFAULT 'Parent',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 3. Academic Records (Enrollment, Program, GPA)
CREATE TABLE IF NOT EXISTS academic_records (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  enrollment_status VARCHAR(50) DEFAULT 'Active',
  academic_program VARCHAR(255),
  department VARCHAR(255),
  semester VARCHAR(50),
  credit_hours INTEGER DEFAULT 0,
  gpa DECIMAL(5, 2) DEFAULT 0.00,
  cgpa DECIMAL(5, 2) DEFAULT 0.00,
```

```sql
-- 4. Attendance Tracking
CREATE TABLE IF NOT EXISTS attendance (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  semester VARCHAR(50),
  classes_attended INTEGER DEFAULT 0,
  total_classes INTEGER DEFAULT 0,
  attendance_percentage DECIMAL(5, 2) DEFAULT 0.00,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 5. Assessment & Examination Results
CREATE TABLE IF NOT EXISTS assessments (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  semester VARCHAR(50),
  internal_marks DECIMAL(5, 2) DEFAULT 0.00,
  quiz_marks DECIMAL(5, 2) DEFAULT 0.00,
  semester_marks DECIMAL(5, 2) DEFAULT 0.00,
  total_marks DECIMAL(5, 2) DEFAULT 0.00,
  grade VARCHAR(5),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 6. Financial & Fee Records
CREATE TABLE IF NOT EXISTS fee_records (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  total_fees DECIMAL(10, 2) DEFAULT 0.00,
  fees_paid DECIMAL(10, 2) DEFAULT 0.00,
  pending_dues DECIMAL(10, 2) DEFAULT 0.00,
  payment_status VARCHAR(50) DEFAULT 'Pending',
  scholarship_amount DECIMAL(10, 2) DEFAULT 0.00,
  scholarship_type VARCHAR(100),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```sql
-- ==============================================
-- Indexes for Performance
-- ==============================================
CREATE INDEX IF NOT EXISTS idx_students_student_id ON students(student_id);
CREATE INDEX IF NOT EXISTS idx_students_name ON students(full_name);
CREATE INDEX IF NOT EXISTS idx_guardians_student ON guardians(student_id);
CREATE INDEX IF NOT EXISTS idx_academic_student ON academic_records(student_id);
CREATE INDEX IF NOT EXISTS idx_academic_department ON academic_records(department);
CREATE INDEX IF NOT EXISTS idx_academic_enrollment ON academic_records(enrollment_status);
CREATE INDEX IF NOT EXISTS idx_attendance_student ON attendance(student_id);
CREATE INDEX IF NOT EXISTS idx_assessments_student ON assessments(student_id);
CREATE INDEX IF NOT EXISTS idx_fee_records_student ON fee_records(student_id);
```

Figure 2.1 depicts the relational schema for the project. The students table serves as the central repository, linked by the unique student_id.

2.2 Description of Tables

Table Name: students **Description:** Stores all information regarding a student, including personal details, academic records, and fee status.

Table 1: students

Attribute	Data Type	Size	Constraints	Description
id	SERIAL	-	PRIMARY KEY	Auto-incrementing ID
student_id	VARCHAR	50	UNIQUE, NOT NULL	Registration ID
full_name	VARCHAR	255	NOT NULL	Full name
date_of_birth	DATE	-	-	Date of birth
gender	VARCHAR	20	-	Gender
address	TEXT	-	-	Address
phone	VARCHAR	20	-	Phone number
email	VARCHAR	255	-	Email address
admission_date	DATE	-	DEFAULT CURRENT_DATE	Admission date
created_at	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	Record created
updated_at	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	Record updated

Table 2: guardians

Attribute	Data Type	Size	Constraints	Description
id	SERIAL	-	PRIMARY KEY	Auto-incrementing ID
student_id	INTEGER	-	FK → students(id), CASCADE	Link to student
guardian_name	VARCHAR	255	NOT NULL	Guardian name
guardian_phone	VARCHAR	20	-	Guardian phone
guardian_email	VARCHAR	255	-	Guardian email
relationship	VARCHAR	50	DEFAULT 'Parent'	Relationship type

Table 3: academic_records

Attribute	Data Type	Size	Constraints	Description
id	SERIAL	-	PRIMARY KEY	Auto-incrementing ID
student_id	INTEGER	-	FK → students(id), CASCADE	Link to student
enrollment_status	VARCHAR	50	DEFAULT 'Active'	Enrollment status
academic_program	VARCHAR	255	-	Program name
department	VARCHAR	255	-	Department
semester	VARCHAR	50	-	Current semester
credit_hours	INTEGER	-	DEFAULT 0	Credit hours
gpa	DECIMAL	5,2	DEFAULT 0.00	GPA
cgpa	DECIMAL	5,2	DEFAULT 0.00	Cumulative GPA

Table 4: attendance

Attribute	Data Type	Size	Constraints	Description
id	SERIAL	-	PRIMARY KEY	Auto-incrementing ID
student_id	INTEGER	-	FK → students(id), CASCADE	Link to student
semester	VARCHAR	50	-	Semester
classes_attended	INTEGER	-	DEFAULT 0	Classes attended
total_classes	INTEGER	-	DEFAULT 0	Total classes
attendance_percentage	DECIMAL	5,2	DEFAULT 0.00	Percentage

Table 5: assessments

Attribute	Data Type	Size	Constraints	Description
id	SERIAL	-	PRIMARY KEY	Auto-incrementing ID
student_id	INTEGER	-	FK → students(id), CASCADE	Link to student
semester	VARCHAR	50	-	Semester
internal_marks	DECIMAL	5,2	DEFAULT 0.00	Internal marks
quiz_marks	DECIMAL	5,2	DEFAULT 0.00	Quiz marks
semester_marks	DECIMAL	5,2	DEFAULT 0.00	Semester exam marks
total_marks	DECIMAL	5,2	DEFAULT 0.00	Total marks
grade	VARCHAR	5	-	Letter grade

Table 6: fee_records

Attribute	Data Type	Size	Constraints	Description
id	**SERIAL**	**-**	**PRIMARY KEY**	**Auto-incrementing ID**
student_id	**INTEGER**	**-**	**FK → students(id), CASCADE**	**Link to student**
total_fees	**DECIMAL**	**10,2**	**DEFAULT 0.00**	**Total fees**
fees_paid	**DECIMAL**	**10,2**	**DEFAULT 0.00**	**Fees paid**
pending_dues	**DECIMAL**	**10,2**	**DEFAULT 0.00**	**Pending dues**
payment_status	**VARCHAR**	**50**	**DEFAULT 'Pending'**	**Payment status**
scholarship_amount	**DECIMAL**	**10,2**	**DEFAULT 0.00**	**Scholarship**
scholarship_type	**VARCHAR**	**100**	**-**	**Scholarship type**

2.3 Creation of Database and Tables - DDL Commands

The Data Definition Language (DDL) commands used to create the database structure are as follows:

Create Table Command:

```
CREATE TABLE IF NOT EXISTS students (
  id SERIAL PRIMARY KEY,
  student_id VARCHAR(50) UNIQUE NOT NULL,
  full_name VARCHAR(255) NOT NULL,
  date_of_birth DATE,
  gender VARCHAR(20),
  address TEXT,
  phone VARCHAR(20),
  email VARCHAR(255),
  admission_date DATE DEFAULT CURRENT_DATE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```

```
);

CREATE TABLE IF NOT EXISTS guardians (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  guardian_name VARCHAR(255) NOT NULL,
  guardian_phone VARCHAR(20),
  guardian_email VARCHAR(255),
  relationship VARCHAR(50) DEFAULT 'Parent',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS academic_records (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  enrollment_status VARCHAR(50) DEFAULT 'Active',
  academic_program VARCHAR(255),
  department VARCHAR(255),
  semester VARCHAR(50),
  credit_hours INTEGER DEFAULT 0,
  gpa DECIMAL(5, 2) DEFAULT 0.00,
  cgpa DECIMAL(5, 2) DEFAULT 0.00,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS attendance (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  semester VARCHAR(50),
  classes_attended INTEGER DEFAULT 0,
  total_classes INTEGER DEFAULT 0,
  attendance_percentage DECIMAL(5, 2) DEFAULT 0.00,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```

```
);

CREATE TABLE IF NOT EXISTS assessments (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  semester VARCHAR(50),
  internal_marks DECIMAL(5, 2) DEFAULT 0.00,
  quiz_marks DECIMAL(5, 2) DEFAULT 0.00,
  semester_marks DECIMAL(5, 2) DEFAULT 0.00,
  total_marks DECIMAL(5, 2) DEFAULT 0.00,
  grade VARCHAR(5),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS fee_records (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  total_fees DECIMAL(10, 2) DEFAULT 0.00,
  fees_paid DECIMAL(10, 2) DEFAULT 0.00,
  pending_dues DECIMAL(10, 2) DEFAULT 0.00,
  payment_status VARCHAR(50) DEFAULT 'Pending',
  scholarship_amount DECIMAL(10, 2) DEFAULT 0.00,
  scholarship_type VARCHAR(100),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Create Index Commands:

```
CREATE INDEX IF NOT EXISTS idx_students_student_id ON students(student_id);
CREATE INDEX IF NOT EXISTS idx_students_name ON students(full_name);
CREATE INDEX IF NOT EXISTS idx_guardians_student ON guardians(student_id);
CREATE INDEX IF NOT EXISTS idx_academic_student ON academic_records(student_id);
CREATE INDEX IF NOT EXISTS idx_academic_department ON academic_records(department);
CREATE INDEX IF NOT EXISTS idx_attendance_student ON attendance(student_id);
CREATE INDEX IF NOT EXISTS idx_assessments_student ON assessments(student_id);
```

CREATE INDEX IF NOT EXISTS idx_fee_records_student ON fee_records(student_id);

2.4 Insertion of Tuples into the Table – DML Commands

Data Manipulation Language (DML) commands are used to insert data into the table.

Example Insertion:

-- *Insert into students*

INSERT INTO students (student_id, full_name, date_of_birth, gender, address, phone, email)

VALUES ('CS2026001', 'Rahul Sharma', '2004-05-15', 'Male', '123 MG Road, Chennai', '9876543210', 'rahul.s@example.com');

-- *Insert guardian*

INSERT INTO guardians (student_id, guardian_name, guardian_phone)

VALUES (1, 'Rajesh Sharma', '9876500001');

-- *Insert academic record*

INSERT INTO academic_records (student_id, enrollment_status, department, semester, gpa, cgpa)

VALUES (1, 'Active', 'Computer Science', '4th', 8.5, 8.2);

-- *Insert attendance*

INSERT INTO attendance (student_id, semester, classes_attended, total_classes, attendance_percentage)

VALUES (1, '4th', 42, 50, 84.00);

-- *Insert assessment*

INSERT INTO assessments (student_id, semester, internal_marks, quiz_marks, semester_marks, total_marks, grade)

VALUES (1, '4th', 18.0, 8.5, 72.0, 98.5, 'A');

-- *Insert fee record*

INSERT INTO fee_records (student_id, total_fees, fees_paid, pending_dues, payment_status)

VALUES (1, 50000.00, 25000.00, 25000.00, 'Pending');

CHAPTER 3

COMPLEX QUERIES BASED ON THE CONCEPTS OF CONSTRAINTS, SETS, JOINS, VIEWS, TRIGGERS AND CURSORS

3.1 Adding Constraints and Queries Based on Constraints

Question: Alter the students table to ensure that the gpa does not exceed 10.00 and the email field is unique, then try to insert a violating record.

SQL Statement:

-- GPA is now in academic_records table

ALTER TABLE academic_records ADD CONSTRAINT check_gpa_range CHECK (gpa <= 10.00);

ALTER TABLE students ADD CONSTRAINT unique_email UNIQUE (email);

-- Attempting check constraint violation

INSERT INTO academic_records (student_id, gpa) VALUES (1, 11.0);

3.2 Queries Based on Aggregate Functions

Question: Find the average GPA, maximum Total Fees paid, and total number of students in the 'Computer Science' department.

SQL Statement:

```
SELECT
    COUNT(*) as total_students,
    ROUND(AVG(ar.gpa), 2) as average_gpa,
    MAX(fr.fees_paid) as max_fees_paid
FROM students s
JOIN academic_records ar ON ar.student_id = s.id
JOIN fee_records fr ON fr.student_id = s.id
WHERE ar.department = 'Computer Science';
```

Output:

total_students	average_gpa	max_fees_paid
45	8.45	50000.00

3.3 Complex Queries Based on Sets

Question: Find the list of students who have either a GPA greater than 9.0 OR have paid their full fees, using UNION.

SQL Statement:

```
SELECT s.full_name, ar.department FROM students s

JOIN academic_records ar ON ar.student_id = s.id WHERE ar.gpa > 9.0

UNION

SELECT s.full_name, ar.department FROM students s

JOIN academic_records ar ON ar.student_id = s.id

JOIN fee_records fr ON fr.student_id = s.id WHERE fr.pending_dues = 0;
```

Output:

full_name	department
Priya Verma	Information Technology
Arjun Das	Computer Science

3.4 Complex Queries Based on Subqueries

Question: Find the details of students whose GPA is higher than the average GPA of their respective department.

SQL Statement:

```
SELECT s.full_name, ar.department, ar.gpa

FROM students s

JOIN academic_records ar ON ar.student_id = s.id

WHERE ar.gpa > (

   SELECT AVG(ar2.gpa)

   FROM academic_records ar2

   WHERE ar2.department = ar.department);
```

Output:

full_name	department	gpa
Priya Verma	Information Technology	9.1
Rahul Sharma	Computer Science	8.5

3.5 Complex Queries Based on Joins (Self-Join)

Question: Find pairs of students who are from the same department and have the same guardian name (siblings).

SQL Statement:

SELECT s1.full_name AS Student1, s2.full_name AS Student2, ar.department, g1.guardian_name

FROM students s1

JOIN guardians g1 ON g1.student_id = s1.id

JOIN guardians g2 ON g1.guardian_name = g2.guardian_name AND g1.student_id != g2.student_id

JOIN students s2 ON s2.id = g2.student_id

JOIN academic_records ar ON ar.student_id = s1.id;

Output:

Student1	Student2	department	guardian_name
Amit Kumar	Sumit Kumar	Electronics	Rajesh Kumar
Sumit Kumar	Amit Kumar	Electronics	Rajesh Kumar

3.6 Complex Queries Based on Views

Question: Create a view named high_achievers for students with a CGPA greater than 9.0 and display them.

SQL Statement:

CREATE VIEW high_achievers AS

SELECT s.student_id, s.full_name, ar.department, ar.cgpa

FROM students s

JOIN academic_records ar ON ar.student_id = s.id

```
WHERE ar.cgpa > 9.0;

SELECT * FROM high_achievers;
```

Output:

student_id	full_name	department	cgpa
CS2026002	Priya Verma	Information Technology	9.0
CS2026005	Sneha Reddy	Computer Science	9.4

3.7 Complex Queries Based on Triggers

Question: Create a trigger to automatically update updated_at timestamp whenever a student record is modified.

SQL Statement:

```
CREATE OR REPLACE FUNCTION update_timestamp_column()

RETURNS TRIGGER AS $$

BEGIN

   NEW.updated_at = CURRENT_TIMESTAMP;

   RETURN NEW;

END;

$$ language 'plpgsql';


CREATE TRIGGER update_student_modtime

   BEFORE UPDATE ON students

   FOR EACH ROW

   EXECUTE PROCEDURE update_timestamp_column();


-- Update a record to test

UPDATE students SET phone = '9988776655' WHERE student_id = 'CS2026001';
```

Output: UPDATE 1 (The updated_at column for Rahul Sharma is updated to the current time).

3.8 Complex Queries Based on Cursors

Question: use a cursor to iterate through all students in the 'Mechanical' department and define a 10% scholarship for them if their GPA is above 8.0.

SQL Statement:

```
DO $$
DECLARE
    student_cursor CURSOR FOR
        SELECT s.id, fr.total_fees, ar.gpa
        FROM students s
        JOIN academic_records ar ON ar.student_id = s.id
        JOIN fee_records fr ON fr.student_id = s.id
        WHERE ar.department = 'Mechanical';
    record RECORD;
BEGIN
    OPEN student_cursor;
    LOOP
        FETCH student_cursor INTO record;
        EXIT WHEN NOT FOUND;
        IF record.gpa > 8.0 THEN
            UPDATE fee_records
            SET scholarship_amount = record.total_fees * 0.10
            WHERE student_id = record.id;
        END IF;
    END LOOP;
    CLOSE student_cursor;
END $$;
```

Output: DO (Query executed successfully, updating scholarship amounts for eligible Mechanical students).

CHAPTER 4

ANALYZING THE PITFALLS, IDENTIFYING THE DEPENDENCIES, AND APPLYING NORMALIZATIONS

4.1 Analyse the Pitfalls in Relations

In the initial design phase, databases often suffer from anomalies due to unnormalized schemas. The common pitfalls include:

- **Redundancy:** Storing the same data multiple times (e.g., repeating department details for every student).
- **Update Anomalies:** If a piece of data changes (e.g., a HOD changes), it must be updated in multiple places, risking inconsistency.
- **Insertion Anomalies:** Inability to insert data because it depends on other missing data (e.g., cannot add a department without a student).
- **Deletion Anomalies:** Deleting a record causing unintended loss of other data (e.g., deleting the last student in a department deletes the department info).

To resolve these, we apply normalization techniques.

4.2 First Normal Form (1NF)

4.2.1 Identify Dependency A relation is in 1NF if it contains only atomic (indivisible) values and there are no repeating groups. *Violation in students table:* The attributes club_memberships and extracurricular_activities allow multiple values stored as comma-separated text (e.g., "Chess, Debate"). This violates atomicity.

4.2.2 Apply Normalization to 1NF To convert to 1NF, we remove the multi-valued attributes and create separate records or tables.

Before 1NF:

student_id	full_name	club_memberships
CS001	Rahul	Chess, Debate
CS002	Priya	Music

After 1NF (Decomposition): We create a separate table Student_Clubs.

Table: Student_Clubs

student_id	club_name
CS001	Chess

student_id	club_name
CS001	Debate
CS002	Music

Violation: In the original unnormalized schema, the students table contained club_memberships and extracurricular_activities as comma-separated text fields. In our normalized 6-table design, these problematic attributes have been removed entirely, and the relevant data is now separated into domain-specific tables.

4.3 Second Normal Form (2NF)

In our implemented 6-table design, we have applied this normalization. The guardians table stores guardian details separately, referenced by student_id FK from the students table.

4.3.1 Identify Dependency A relation is in 2NF if it is in 1NF and has no **partial dependencies** (i.e., non-prime attributes must depend on the full primary key). *Scenario:* Consider a hypothetical Student_Marks table with Composite Key (student_id, subject_code). Attributes: student_name, marks. Dependency: student_name depends only on student_id, not subject_code. This is a partial dependency.

4.3.2 Apply Normalization to 2NF We split the table to separate student details from their marks.

Before 2NF:

student_id	subject_code	student_name	marks
CS001	CSE101	Rahul	85
CS001	CSE102	Rahul	90

After 2NF: **Table 1: Students (Attributes depend on student_id)**

student_id	student_name
CS001	Rahul

Table 2: Marks (Attributes depend on student_id + subject_code)

student_id	subject_code	marks
CS001	CSE101	85

student_id	subject_code	marks
CS001	CSE102	90

4.4 Third Normal Form (3NF)

4.4.1 Identify Dependency A relation is in 3NF if it is in 2NF and has no **transitive dependencies** (non-prime attribute depends on another non-prime attribute). *Violation in students table:* The table contains guardian_name, guardian_phone, and guardian_email. Dependency: guardian_phone depends on guardian_name (or a Guardian ID) rather than directly on the student_id. If two students have the same guardian (siblings), the guardian's phone is repeated.

4.4.2 Apply Normalization to 3NF We move guardian details to a separate Guardians table.

Before 3NF:

student_id	guardian_name	guardian_phone
CS001	Rajesh Kumar	9876543210
CS002	Rajesh Kumar	9876543210

After 3NF: **Table 1: Guardians**

guardian_id	guardian_name	guardian_phone
G01	Rajesh Kumar	9876543210

Table 2: Students

student_id	guardian_id
CS001	G01
CS002	G01

4.5 Boyce-Codd Normal Form (BCNF)

4.5.1 Identify Dependency BCNF is a stricter version of 3NF. A table is in BCNF if for every functional dependency X -> Y, X is a superkey. *Scenario:* A Student_Project table where (student_id, project_name) -> guide_name. But also guide_name -> project_name (if a guide only leads one project). Here guide_name is not a superkey but determines part of the PK.

4.5.2 Apply Normalization to BCNF Split into Guide_Projects and Student_Guide.

After BCNF: **Table: Guide_Projects**

guide_name	project_name
Dr. Smith	AI Chatbot

Table: Student_Guide

student_id	guide_name
CS001	Dr. Smith

4.6 Fourth Normal Form (4NF)

4.6.1 Identify Dependency 4NF deals with **Multi-Valued Dependencies (MVD)**. A table is in 4NF if it is in BCNF and has no MVDs. *Scenario:* A student has multiple Hobbies and multiple Certifications. student_id ->> Hobby student_id ->> Certification If we store them in one table (student_id, hobby, certification), we get a Cartesian product (every hobby paired with every certification), which is redundant.

4.6.2 Apply Normalization to 4NF Decompose into two separate tables.

After 4NF: **Table: Student_Hobbies**

student_id	hobby
CS001	Chess

Table: Student_Certifications

student_id	certification
CS001	AWS

4.7 Fifth Normal Form (5NF)

4.7.1 Identify Dependency 5NF (Project-Join Normal Form) deals with cases where a relation can be reconstructed from smaller relations but not all combinations are valid (Join Dependency). It ensures that no information is lost or spurious records created when joining decomposed tables. *Scenario:* A ternary relationship between Student, Subject, and Professor, where:

- Student takes Subject.
- Professor teaches Subject.
- Student learns from Professor.

4.7.2 Apply Normalization to 5NF If the relationship is cyclic but cannot be decomposed without losing the specific "Student X learns Subject Y from Professor Z" fact, it is already in

5NF. If independent, decompose into Student_Subject, Subject_Professor, and Student_Professor.

The schema is normalized up to BCNF for core entities to ensure strong data integrity while maintaining practical query complexity.

CHAPTER 5

Lock Mode	Description
ROW SHARE	Allows concurrent access; prevents other sessions from locking the table exclusively.
ROW EXCLUSIVE	Prevents other sessions from locking in share mode. Used by default for DML.
SHARE	Allows queries but not updates or deletes.
SHARE ROW EXCLUSIVE	A mix; more restrictive than SHARE.
EXCLUSIVE	Prevents all other access — full table lock.

IMPLEMENTATION OF CONCURRENCY CONTROL AND RECOVERY MECHANISMS

5.1 Introduction to Transactions

A transaction is a logical unit of work that contains one or more SQL statements. It is atomic in nature: either all operations are committed, or all are rolled back.

5.1.1 Properties (ACID)

- Atomicity: All statements in the transaction succeed together, or none are applied.
- Consistency: Every committed transaction preserves all database constraints.
- Isolation: Concurrent transactions do not interfere in a way that breaks correctness.
- Durability: Once committed, changes survive crashes and restarts.

5.1.2 Transaction States

A transaction passes through these states:

- Active: Transaction is currently executing.
- Partially Committed: Last statement executed, commit processing not fully finished.
- Failed: Execution cannot continue due to error or conflict.

- Aborted: All effects rolled back.

- Committed: All effects made permanent.

5.1.3 Transaction State Diagram

```
                +----------------+
                |    Active      |
                +----------------+
                      |      |
                | success | failure
                        v      v
+---------------------+  +----------------+
| Partially Committed |  |    Failed      |
+---------------------+  +----------------+
                 |              |
                 | commit done      | rollback
                 v              v
    +---------------+   +---------------+
    |  Committed  |   |  Aborted   |
    +---------------+   +---------------+
```

5.2 Transaction Control Language (TCL)

TCL commands control transaction boundaries and recovery behavior.

5.2.1 Savepoint

A savepoint marks a rollback point inside an active transaction.

`SAVEPOINT savepoint_name;`

5.2.2 Commit

`COMMIT;` permanently stores all changes made in the current transaction.

5.2.3 Rollback

`ROLLBACK;` cancels uncommitted changes. `ROLLBACK TO savepoint_name;` reverts up to a selected point.

5.3 Transaction Examples

5.3.1 Transaction 1: Enroll a Student

```sql
BEGIN;
INSERT INTO students (name, email, phone, dob, gender)
  VALUES ('Rahul', 'rahul@srm.edu', '9876543210', '2004-01-01', 'M')
  RETURNING student_id;

INSERT INTO enrollments (student_id, program_id, year, semester, section)
  VALUES (1, 2, 2024, '4th', 'A');
COMMIT;
```

5.3.2 Transaction 2: Grade Update with SAVEPOINT

```sql
BEGIN;
UPDATE grades SET marks = 85 WHERE grade_id = 1;
SAVEPOINT grade_updated;
UPDATE grades SET grade_letter = 'A' WHERE grade_id = 1;
-- if recalculation fails:
```

```
ROLLBACK TO grade_updated;
COMMIT;
```

5.3.3 Transaction 3: Bulk Grade Update for a Semester

```sql
BEGIN;
UPDATE grades SET marks = 90, grade_letter = 'O'
  WHERE enrollment_id = 1 AND course_id = 3;
UPDATE grades SET marks = 72, grade_letter = 'B'
  WHERE enrollment_id = 1 AND course_id = 4;
COMMIT;
```

5.3.4 Transaction 4: Scholarship Adjustment with Consistent Fee Recalculation

```sql
BEGIN;
UPDATE students SET scholarship = 5000 WHERE student_id = 1;
SAVEPOINT scholarship_applied;
UPDATE students SET pending_fees = total_fees - fees_paid - scholarship
  WHERE student_id = 1;
COMMIT;
```

5.3.5 Transaction 5: Section Transfer with Recovery Option

```sql
BEGIN;
UPDATE enrollments
```

```
SET section = 'B'
WHERE enrollment_id = 1;

SAVEPOINT section_changed;

UPDATE enrollments
SET semester = '5th'
WHERE enrollment_id = 1;

-- if verification fails:
-- ROLLBACK TO section_changed;

COMMIT;
```

5.4 Concurrency Control

Concurrency control ensures correctness under simultaneous transactions.

5.4.1 Locking Demonstrations

a) Table-level shared lock demo:

```sql
BEGIN;
LOCK TABLE students IN SHARE MODE;
SELECT COUNT(*) FROM students;
COMMIT;
```

b) Row-level exclusive lock demo:

```sql
BEGIN;
SELECT * FROM grades WHERE grade_id = 1 FOR UPDATE;
UPDATE grades SET marks = 85 WHERE grade_id = 1;
COMMIT;
```

5.4.2 Lock Compatibility Matrix

	S-lock requested	X-lock requested
S-lock held	Allowed	Blocked
X-lock held	Blocked	Blocked

5.4.3 Concurrency Example (Lost Update Prevention)

Session 1:

```sql
BEGIN;
SELECT * FROM grades WHERE grade_id = 1 FOR UPDATE;  -- row-level X lock
UPDATE grades SET marks = 88 WHERE grade_id = 1;
COMMIT;  -- lock released
```

Session 2:

```sql
BEGIN;
UPDATE grades SET marks = 75 WHERE grade_id = 1;
```

```
-- BLOCKED until Session 1 commits

COMMIT;
```

5.5 Two-Phase Locking (2PL)

Two-Phase Locking is a standard protocol used to guarantee serializable schedules. In the growing phase, a transaction acquires all required locks and does not release any lock. In the shrinking phase, it releases locks and cannot acquire new ones. This prevents many anomalies (including lost updates) and is the conceptual basis of strict locking behavior demonstrated in the `FOR UPDATE` examples above.

CHAPTER 6

FRONT-END AND BACK-END CODE OF STUDENT DATABASE MANAGEMENT SYSTEM

6.1 Front–End Module Codes

File: app/page.tsx (Main Dashboard) This component renders the landing page with options to Enter Data or View Students.

```
'use client';


import { useState, useEffect } from 'react';

import Link from 'next/link';


export default function HomePage() {

  const [isDark, setIsDark] = useState(true);

  const [mounted, setMounted] = useState(false);


  useEffect(() => {

    setMounted(true);

    const saved = localStorage.getItem('theme');

    if (saved === 'light') {

      setIsDark(false);

      document.body.classList.add('light');
```

```
  }
}, []);

// ... (Theme toggle logic)

if (!mounted) return null;

return (
  <div style={{ minHeight: '100vh', display: 'flex', flexDirection: 'column' }}>
    {/* Header Section */}
    <header className="animate-fade-in" style={{ /* styles */ }}>
      {/* ... */}
      <h1>Student System</h1>
    </header>

    {/* Main Content */}
    <main style={{ flex: 1, display: 'flex', alignItems: 'center', justifyContent: 'center' }}>
      <div style={{ maxWidth: '800px', width: '100%', textAlign: 'center' }}>
        <h2>Student Database<br />Management System</h2>

        {/* Navigation Cards */}
        <div className="animate-slide-up delay-100" style={{ display: 'grid', gap: '24px' }}>
          <Link href="/enter">
            <div className="card">
              <h3>Enter Data</h3>
              <p>Add new student records</p>
            </div>
          </Link>
```

```
          <Link href="/view">
            <div className="card">
             <h3>View Students</h3>
             <p>Browse and manage database</p>
            </div>
          </Link>
        </div>
      </div>
    </main>
  </div>
 );
}
```

File: app/view/ViewClient.tsx (Student List View) Displays the list of students with search, filter, and delete functionality.

```
'use client';

import { useState, useEffect } from 'react';
import { deleteStudent } from '@/lib/actions/students';
import { StudentFull } from '@/lib/types';

export default function ViewClient({ initialStudents }: ViewClientProps) {
  const [students, setStudents] = useState<StudentFull[]>(initialStudents);
  const [searchTerm, setSearchTerm] = useState('');

  const filteredStudents = students.filter(student =>
    student.full_name.toLowerCase().includes(searchTerm.toLowerCase()) ||
    student.student_id.toLowerCase().includes(searchTerm.toLowerCase())
  );
```

```
const handleDelete = async (id: number) => {
    if (!confirm('Are you sure you want to delete this student?')) return;
    const result = await deleteStudent(id);
    if (result.success) {
        setStudents(students.filter(s => s.id !== id));
    }
};

return (
    <div className="animate-fade-in">
        {/* Search Bar */}
        <input
            type="text"
            placeholder="Search by name, ID..."
            value={searchTerm}
            onChange={(e) => setSearchTerm(e.target.value)}
        />

        {/* Student List */}
        {filteredStudents.map((student) => (
            <div key={student.id} className="card">
                <h3>{student.full_name}</h3>
                <p>Dept: {student.department}</p>
                <p>GPA: {student.gpa}</p>
                <button onClick={() => handleDelete(student.id)}>Delete</button>
            </div>
        ))}
    </div>
);
```

```
}
```

6.2 Database Connectivity

File: lib/db/index.ts (Database Connection) Establishes the connection pool to the PostgreSQL database.

```
import { Pool, QueryResult, QueryResultRow } from 'pg';


if (!process.env.DATABASE_URL) {
    throw new Error('DATABASE_URL is not defined');
}


// Create a connection pool
export const pool = new Pool({
    connectionString: process.env.DATABASE_URL,
    ssl: { rejectUnauthorized: false },
    max: 20,
    idleTimeoutMillis: 30000,
});


// Helper function for queries
export async function query<T extends QueryResultRow = any>(
    text: string,
    params?: any[]
): Promise<QueryResult<T>> {
    const start = Date.now();
    try {
        const result = await pool.query<T>(text, params);
        return result;
    } catch (error) {
        console.error('Database query error:', error);
```

```
        throw error;
    }
}
```

File: lib/actions/students.ts (Backend Logic) Server actions for CRUD operations on student records.

```
'use server';

import { query } from '@/lib/db';
import { revalidatePath } from 'next/cache';

// Get all students (6-table JOIN)
export async function getAllStudents() {
    const result = await query(`
        SELECT s.*, g.guardian_name, g.guardian_phone, g.guardian_email,
            ar.enrollment_status, ar.department, ar.semester, ar.gpa, ar.cgpa,
            att.attendance_percentage, att.classes_attended, att.total_classes,
            a.internal_marks, a.quiz_marks, a.semester_marks, a.total_marks, a.grade,
            f.total_fees, f.fees_paid, f.pending_dues, f.payment_status
        FROM students s
        LEFT JOIN guardians g ON g.student_id = s.id
        LEFT JOIN academic_records ar ON ar.student_id = s.id
        LEFT JOIN attendance att ON att.student_id = s.id
        LEFT JOIN assessments a ON a.student_id = s.id
        LEFT JOIN fee_records f ON f.student_id = s.id
        ORDER BY s.created_at DESC
    `);
    return { success: true, data: result.rows };
}
```

```
// Create student (transactional multi-table insert)
export async function createStudent(data: any) {
    const client = await getClient();
    await client.query('BEGIN');
    // INSERT into students → get id → INSERT into 5 child tables
    await client.query('COMMIT');
}
// Create new student
export async function createStudent(data: any) {
    try {
        const result = await query(
            `INSERT INTO students (student_id, full_name, department, gpa, ...)
             VALUES ($1, $2, $3, $4, ...) RETURNING *`,
            [data.student_id, data.full_name, data.department, data.gpa]
        );
        revalidatePath('/');
        return { success: true, data: result.rows[0] };
    } catch (error: any) {
        return { success: false, error: error.message };
    }
}
```

CHAPTER 7

RESULTS AND DISCUSSIONS

7.1 Screenshots of Front-end

Figure 7.1: Home Page Dashboard

The home page serves as the entry point for the application, offering a modern and intuitive interface. Users can navigate to "Enter Data" to add new student records or "View Students" to manage existing data. The design features a responsive layout with a toggle for dark/light mode.

Figure 7.2: Student List View

The student list view displays all registered students in a card-based layout. It includes a search bar for filtering students by name, ID, or department. Each student card can be expanded to view detailed information such as academic performance, attendance, and fee status.

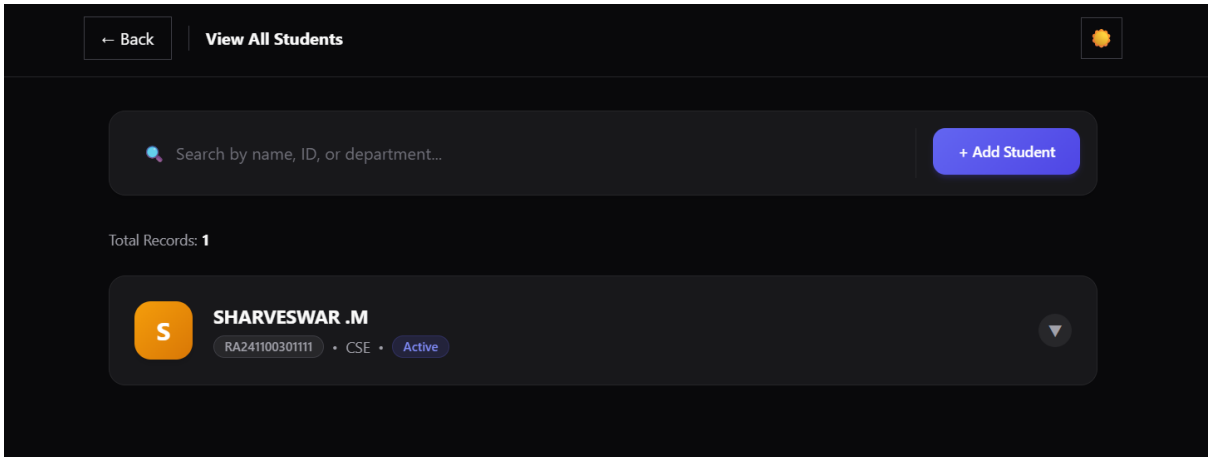


Figure 7.3: Student Detail Expansion

Clicking on a student card reveals comprehensive details. This modular design ensures that the interface remains clean while providing easy access to in-depth data.

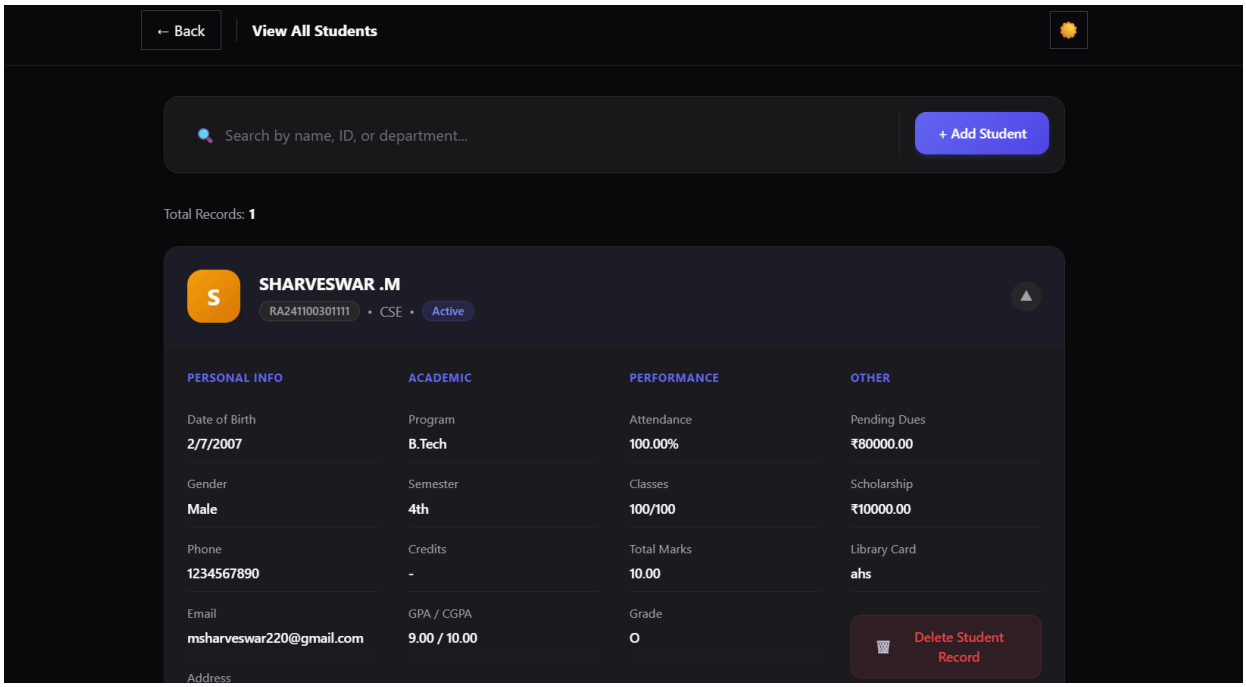
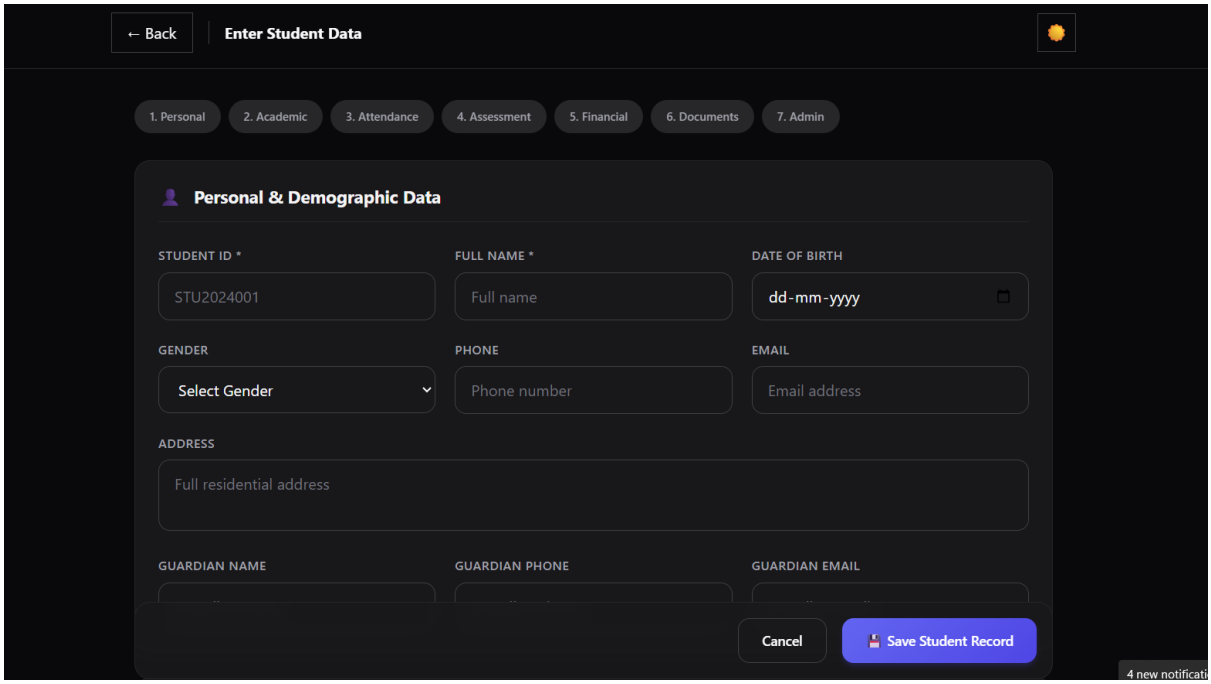


Figure 7.4: Add Student Form

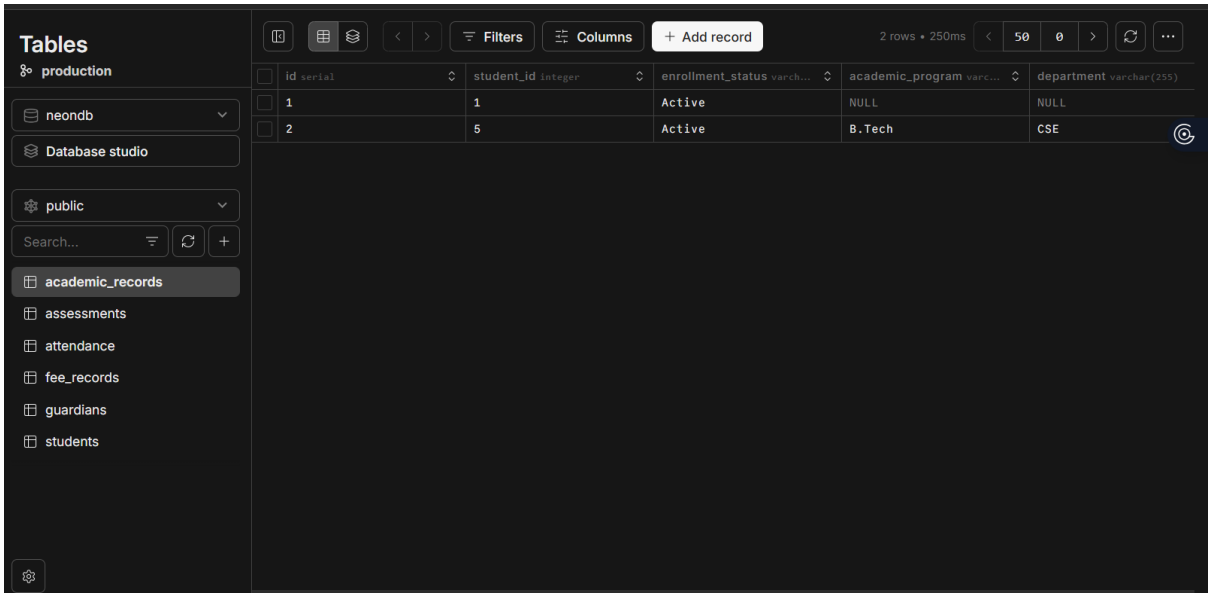
The data entry form allows administrators to input new student details. It includes validation checks to ensure data integrity (e.g., preventing negative values for marks or fees).



7.2 Screenshots of Database

Figure 7.5: Database Connection

The application connects to a PostgreSQL database hosted on Neon. The connection is secured using SSL, ensuring data privacy during transmission.



7.3 Discussion

The implementation of the Student Database Management System successfully addresses the problems identified in the manual system.

- **Efficiency:** Data retrieval time has been significantly reduced.
- **Accuracy:** Automated calculations for GPA and CGPA eliminate manual errors.
- **Usability:** The modern user interface requires minimal training for administrators.
- **Scalability:** The use of PostgreSQL ensures the system can handle a growing dataset without performance issues.

Future enhancements could include role-based access control (RBAC) to differentiate between admin and student access, and a module for generating PDF reports directly from the system.

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